

prospective situations as lightning-fast yet super-small computers, invisibility cloaks, smartphones that last weeks between charges and computers that we can fold up and carry in our pockets wherever we go.

The range of applications for graphene in 2D materials in electronics extends far beyond just digital logic. While much research effort is devoted to seeing if graphene can replace silicon as the basis for the next generation of computer chips, this is not its only potential application in the broader field of electronics. In addition to its use in transistors for digital logic, graphene is also being investigated for use in flexible electronics where it can enable conductive inks and serve as a replacement for indium-tin-oxide as transparent conductors in these devices.

There are also the more workhorse roles as an alternative material for interconnects and heat-sinks. Graphene research is focused on applications like energy, membranes, composites and coatings, biomedical, sensors and electronics. This is only the start and these are only the first steps. Potential of graphene is limited only by imagination.

Electronics and energy storage could also be revolutionised by graphene. Flexible, durable, semi-transparent mobile phones, wearable technology, clothing that communicates, electric sports cars and lightweight planes are some of the future technologies that are becoming realistic in our present. Hence, there are a number of growing applications of graphene based technology, but this article is limited to introducing briefly the applications of graphene in electronics.

Graphene based flexible displays. One particular area in which you will soon begin to see graphene used on a commercial scale

is in optoelectronics, specifically touchscreens, liquid crystal displays (LCDs) and organic light emitting diodes (OLEDs). For a material to be able to be used in optoelectronic applications, it must be able to transmit more than 90 per cent of light and also offer exceptional electrical conductive properties and therefore low electrical resistance.

Graphene is an almost completely-transparent material and is able to optically transmit up to 97.7 per cent of light. It is also highly conductive and so it would work very well in optoelectronic applications such as LCD touchscreens for smartphones, tablets, desktop computers and televisions.

Researchers have developed a method for producing graphene-treated silver nanowires, which could significantly reduce production costs for nanowire based displays. This could mean that graphene offers a real alternative to indium-tin-oxide in flexible low-cost touchscreen displays. The work has cut the amount of expensive nanowires required to build such touchscreens by more than 50 times besides simplifying the production process by using graphene.

The key to the new process is to exploit graphene's unmatched conducting properties. To do this, researchers changed from using graphene-oxide that is typically used in a solution based process to create nanowires to pristine graphene. Since pristine graphene is free of oxygen functional groups found in graphene-oxide, it could conduct electricity without any further chemical treatment, which resulted in more than a 50-fold reduction in the number of nanowires needed to produce viable electronic electrodes. This is a real alternative to indium-tin-oxide displays and could replace existing touchscreen technologies in electronic devices.

Energy-efficient transistors. When graphene is placed on

top of boron-nitride, it creates a super-lattice, which is a structure made of aligned, alternating layers of various nanomaterials. The super-lattice can move electrons perpendicular to the electric field without the influence of a magnetic field. This could lead to a new kind of energy-efficient transistor. While the research team has not attempted to build a transistor based on the phenomenon, the super-lattice material has displayed high sensitivity to gate voltage that operates transistors.

Another implication of this discovery is that electrons in the super-lattice appear to behave just like neutrinos, which are mass-less particles that do not interact with most kinds of matter. Researchers believe that this discovery could contribute to our understanding of the Universe.

Random access memory (RAM). Researchers have developed a way of using graphene in order to improve the ferroelectric-tunnel-junction that is a component of RAM. They have improved the ferroelectric-tunnel-junction by combining graphene with ammonia so that it is capable of switching on and off the flow of electrons more completely.

The result is a distinct improvement in the reliability of RAM devices and the ability to read data without having to rewrite it. This is one of the most important differences between previous technology that has already been commercialised and this emergent ferroelectric technology.

In a typical ferroelectric-tunnel-junction design, a ferroelectric layer is placed between two electrodes so that when an electric field is applied to these, direction of the junction's polarisation is reversed. This reversal of polarisation changes the alignment of positive and negative charges, which correspond to zero and one in binary computing.